
Listening to Yourself through the Otherself: On *Background Noise Study* and other works

AGOSTINO DI SCIPIO

Via Salaria Antica Est 33a, 67100 L'Aquila, ITALY
E-mail: discipio@tin.it

In this paper I describe a work of mine, *Background Noise Study*, as a starting point to raise questions of broader relevance for current artistic practices. Issues are presented in terms varying from technical and specific, to more philosophical and general (system-theory, biocybernetics, ecology). Excursions are made into the music-theoretical (understanding of key issues in the sound arts, such as 'space', 'performance', 'form' or 'timbre') and the political (politics of sound, biopolitics of performance and listening).

1. INTRODUCTION

Background Noise Study is the third in a series of works entitled *Audible Ecosystemics*. I composed it during a DAAD residency (Berlin 2004–05), mostly using my own equipment. Technical refinements were pursued in the Elektronisches Studio of the Technische Universität. The premiere took place in the Tesla/Podewils'chen Palais, in Berlin, 5 July 2005. A variant work, *Background Noise Study, with Mouth Performance* (a.k.a. *Background Noise Study, in the Vocal Tract*), was premiered in the same concert. The two can be presented either together or separately. When presented together, the 'mouth performer' enters the performance space shifting silently into the sonic environment developed from the first work, so the performance of the second work starts smoothly after the first, with no break.

In the present paper I would like to focus on *Background Noise Study* as a starting point to discuss themes of relevance to contemporary artistic practice. Where appropriate, observations on the variant work are included.

2. DEFINITIONS

When I am asked to define the *Audible Ecosystemics* works, I have trouble in answering in a way that is both clear to my interlocutors and honest to the particularity of such works. *Background Noise Study* can be defined perhaps as a live-electronics solo performance. That implies the notion of a concert piece. Sometimes, though, it has been presented as a sound installation.¹ Following Schröder (2008), the

Audible Ecosystemics works reflect a tendency in recent sound-art practices where artistic concepts are developed both as concert works and as installations in galleries and other venues.

Feedback Study (2003) is slightly different. Born as a live-electronics solo, it was set up as a sound installation during a workshop in the Alte Mensa of the Johannes Gutenberg University, in Mainz (summer 2004). In that case, however, four performers were to enter the installation and act upon it, creating small resonant niches around a number of microphones standing in the room. That turned the work into a kind of 'performed installation'. The same is noted in (De La Motte 2008) with reference to the *Impulse Response Study* (2002).

Modes of interference n.3 (autonomous feedback network with three or more electric guitars, combo amps, and computer, 2007) was initially designed to be presented as a sound installation. However, Stefan Prins and the ZWERM electric guitar ensemble devised ways to include it in concert presentations they have recently given across Europe. At the time of writing, I am working on *Modes of interference n.4*, a feedback network made of four disassembled guitar amps and other recycled analogue electronics. The plan is to show it as a small installation in galleries, but I could also turn it into a kind of idiosyncratic instrument in performance situations.

I think this kind of confusion as to the manner of presentation of a live sound work is beneficial. It suggests that behind the artistic concept hides a complex and flexible (or 'undecided') relationship to individual and collective experience, reflecting less a fixed temporal constellation, and more a shared, lived experience of time and space; the here-and-now of the presentation.

I view these works as a computation activity immersed in the real world: a *computational mechanism having physical terminals (electroacoustic transducers) reaching into the surrounding environment, constantly affecting the sonic ambience in that environment and constantly being affected by it*. This definition (elaborated in Di Scipio 2008, where *Feedback Study* is more particularly described) evokes both the operational and the systemic element peculiar to the approach; it situates

¹See, for example, <http://emergentsonorities.tumblr.com>.

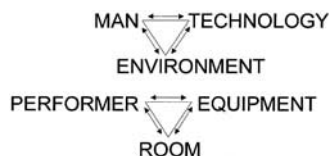


Figure 1.

the technological infrastructure and the very process specific to the work in the real, material space, pointing to a view where the performance process is tightly coupled with the space housing the performance. The basic idea is to address, in the medium of sound, the essential interaction between human agency (performer), technologies (audio equipment) and environment (room space, site) (Figure 1).

I evoke this kind of ecosystemic view because I believe that today it represents a broad and rich cultural framework that can very vividly be experienced in the medium of sound. Our work as composers or sound artists can be understood as the design and experimentation of *performance ecosystem* (Waters 2007). As such, it may foster an understanding of sound and related artistic practices of a kind that possibly goes beyond the technicalities and the aesthetics many artists limit themselves to dealing with. The approach can bridge the creative exploration of sound and auditory perception to questions of social and political relevance in a very tangible way. It helps setting the ground for a 'politics of sound' (or biopolitics of music). This represents today a much needed and plausible cultural agenda where critical discourse and artistic practices directly address the context of, and the means for, desirable sound-producing human actions.

In a performance of *Background Noise Study*, everything is connected to every other thing in the medium of sound: we hear the interactions, the links and clashes sound is born of. No thing capable of vibrating, conveying and channelling sound remains foreign to the overall process. No listener remains an entirely detached, neutral observer. We will touch on such points later. Let us try first to describe a little of how it works.

3. ONTOLOGY (SOUND, SYNCHRONIC EMERGENCE)

Two microphones stand somewhere in the room, far from each other, oriented towards possible noise sources (either internal or external to the room). A 'noise source' here is any physical cause that adds to the room background noise. (Some equipment could also be considered a possible noise source.) If no noise source can be easily determined when preparing for a performance, then the microphones could be positioned in the middle of the room, raised or resting

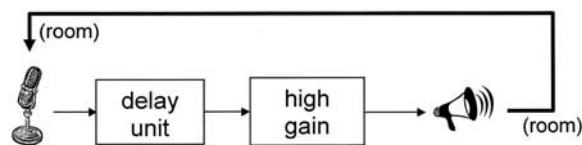


Figure 2.

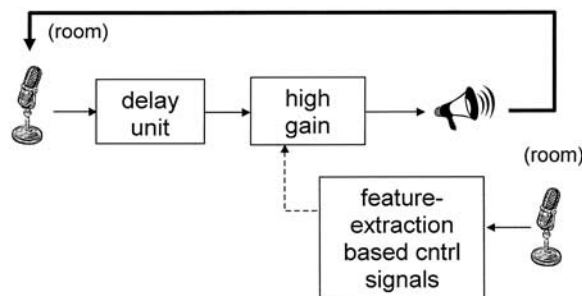


Figure 3.

on the ground, not too far from where human beings sit or stand.

Loudspeakers stand around the room, close to the walls. Their cones face the wall surfaces, not the audience. As many of them can be used as seems appropriate, depending on the room's size and reverberant character.

The microphone sound is routed to a delay unit, with a delay time of 20 seconds (the delay unit and all signal processing mentioned in these pages are implemented as real-time computer applications²). The delayed sound is hugely amplified and heard through the speakers, louder than the natural background noise. It thus recirculates through the room back into the microphone (Figure 2).

The level of the delayed sound is automatically controlled in the computer, with a kind of adaptive peak-dampening mechanism: the output level is constantly driven as an inverse function of the input level. That prevents the sound recirculating in the feedback loop from accumulating to a point of saturation. As a side effect, it introduces amplitude and phase modulations which in turn elicit room resonances other than the strongest, most prominent ones. The attempt to get a variety of room responses using few and simple means is crucial for the performance economy of the work described here (Figure 3).

Notice that the background noise, and all sound the microphone captures in the room, represents not only an *audio signal* but a source of information concern-

²All digital signal processing in the *Audible Ecosystemics* project was developed with Symbolic Sound's KYMA workstation (currently using the Pacarana DSP box). In other works, PURE DATA was also used, although mainly for less expensive computational tasks.

ing the room environment. Based on the real-time extraction of psychoacoustically relevant sonic features, we generate *control signals* driving or modifying the audio signal (its amplitude, in the case just described). In the present context, *audio signal processing* is not at all more crucial than *control signal processing* (Di Scipio 2003). Observed features of sound drive the way sound itself is handled. This idea is pervasive in the technicalities of *Background Noise Study*, and provides the basis for the sounding results to be shaped and articulated in time (see next section). Sound is experienced not only as an energetic potential (*what* acoustical sources and events are there to work with?), but an informational potential too (*how* events should be dealt with, given the properties we observe in them?).

The room's delayed background noise is likely to be perceived, at the beginning of the performance, as the hiss of a somewhat cheap, low-fi electroacoustic set-up. Listeners are deliberately presented with something usually foreign to all music, the undesired trace of the equipment amplifying the ambience sound, magnifying something we usually do not (want to) pay attention to. Ambient noise is always there in the background, yet always before our ears. It typically remains something never listened to, never acknowledged as having a potential for meaning – except when it gets too loud and annoying – in other words, potentially destructive of intended musical messages. In *Background Noise Study* this marginal sound is taken as the only chance for some music to eventually emerge, in a gesture that opens to a particular phenomenology of noise and silence (Meric 2008; Di Scipio 2005b). We start with this 'nothing-musical' and turn it into the basic resource for building something that may be called music.

By way of repeatedly feeding this nothing-musical back into the microphone, a less feeble and richer sonority emerges. It is denser and spectrally more focused, perhaps even pitched, featuring one or more resonant frequencies in its spectrum. Thus far the process replicates Alvin Lucier's *I am sitting in a room* (1970) (Lucier 1995): it uses the room as a resonant filter within a delayed feedback loop. However, in contrast to Lucier's masterwork, *Background Noise Study* has no special event of sound to introduce in the feedback loop. Moreover, through the peak-dampening mechanism, the level is constantly and automatically regulated against the room's own response to the sound of the feedback loop itself. The room is not only a filter, but also a source of information and self-regulation. Overall, the process becomes a sound generation unit, making sure that some sound born 'of nothing' will eventually be there instead of 'silence' (= unnoticed ambience).

The timbral properties of the emergent sonority are determined by the acoustics of the room, but also to a

large extent by the technical characteristics of the electroacoustic transducers involved, their position relative to room surfaces, and of other stages in the electroacoustic chain (pre-amplifiers, mixer routing, etc.). All system components are *individually* relevant. However, and perhaps more importantly, they are *mutually* determinant: the direct and indirect inter-connections among them are more decisive than the single, separate functions. For example, the total gain in the electroacoustic chain is usually handled as a free variable, only relative to a subjective view of 'how loud the music should be'; however, in this case, the total system gain is bound to depend on the acoustics of the given room, and especially on its reverberant character. You can't regulate the level at your will, as if it were not a component feature in a larger, integrated system. Wanting to make it louder then necessary simply does not work.

The delayed feedback loop is active throughout the performance and incessantly feeds the higher-level signal transformations involved (see next section). It keeps the total sound stream alive, nourishing it with all sorts of accidental events happening in the room. *Background Noise Study* welcomes noise and accidental events, and tries to do something with them.

4. MORPHOLOGY (GESTURE, DIACHRONIC EMERGENCE)

The two microphones may be either standing in different niches of the room (a balcony, a corner, etc.), or in the performance area. Besides very tiny sound events born in close proximity, the two capture the *same* sound events. However, they clearly provide *different* signals, depending on the specific position, the distance of reflective surfaces, the phase differences (shifts in time and spatial orientation) of sound waves travelling in the room. In the computer, the microphone sound is subject to amplitude followers and other tracking methods (including some spectral properties). When the sound level in the mic exceeds a given threshold relative to the particular room, the delayed feedback loop described above is temporarily discontinued and gives way to a process that resamples the feedback loop sound from an internal buffer at different speeds. Three such sample-and-playback algorithms are active. They add new materials on top of the delayed feedback loop sound layer, scattered among various frequency regions. The frequency shift factors are also modulated by control data extracted from the input sound. For instance, if the total sound in the room gets richer in higher frequencies, the frequency shift factors get lower, so deeper sounds will in short compensate for the higher frequencies. Overall, the sample-and-playback mechanism takes what is emerging from the feedback loop and

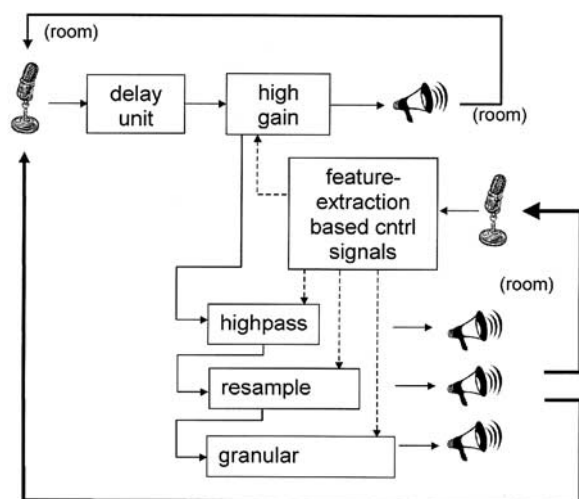


Figure 4.

articulates, enriches or masks it, adding a variety of textural and/or gestural events.

More control-signals are generated based on features observed in the microphone input, and are usually utilised following a kind of compensation principle largely adopted throughout the network of signal-level interactions. For example, built in the basic feedback loop is a high-pass filter, the cut-off frequency of which is made to get higher when the room response remains low and dark for longer stretches of time. As a first consequence, soon the focus in the spectrum will move away from the lower frequency range. As a second consequence, less deep or even shrill sounds will in general elicit weaker room responses. As a third consequence, weaker responses will in turn retard the next threshold-crossing event.

The sound material born of the sample-and-playback mechanisms is not only heard through the speakers, but is also written into a second memory buffer. A ‘granulator’ – a special sampler working on very small chunks of the sample stream – segments and randomises the signal written in the buffer. Grain density and other details are driven by data tracked from the microphone input. The granular process vaporises the sample-and-playback material, turning it into a thinner dust of sound. Overall, it adds micro-level sonic variations (sometimes rhythmical), as well as thicker or thinner statistical textures.

Notice the cascaded signal processing design (shown in Figure 4): (1) delayed feedback loop (with internal peak-dampening and highpass filter) → (2) sample-and-playback → (3) granulation. Each layer provides materials to the next (and to the speakers too). Because of the inherent feedback chain, materials generated with step 2 and step 3 and heard in the room are also circulated in the main delayed feedback loop. Thus the sampler sound and the granulation

sound will not only interfere with the delayed feedback loop, but through that loop they will actually provide themselves with materials to be re-sampled and re-granulated, establishing a kind of iterative transformation (an iterated nonlinear process).

The materials arising from the iterated processing are likely to cause stronger and spectrally richer room responses. The total sound in the room could then increase and stay relatively loud for a prolonged time. If it stays loud for 20 seconds or longer, the amplitude threshold against which the microphone levels are measured will be continually or repeatedly exceeded. Eventually, this causes the feedback loop to be discontinued over a prolonged time span, and thus provide no more sound to the other processes for a period. In other words, when the total room sound becomes loud enough and stays loud for too long a time, the total process automatically shuts itself down and settles.

The latter event is heard as a dramatical and sudden gesture, a remarkable moment in the unfolding of the performance. Suddenly the process resorts to silence and stays silent for around 20 seconds – just like at the beginning. Then, the room background noise is heard again in the speakers, and the process starts again.

A crisis (system grows to a point of saturation) opens to a fresh restart (silence, reset of system functionalities). However, it can leave traces, leftover materials (especially sound materials recirculating in the buffers of the samplers and the granular processing). Therefore, the silence restored will be punctuated with sound wastes adding to the room’s own background noise. Also, the ambience in the room could have changed (new listeners have come in, some have moved from one seat to another, a horn is heard in the distance, a baby cries just behind the door, etc.). In short, the new starting conditions are not really the same as in the beginning of the performance, so going through the entire process again is not like telling the same story again, although the overall dynamic profile could be of a similar kind. New materials will possibly arise because of new interactions among the system components, and because of changes in the system components themselves.

In Figure 5, a map of the main nodes in the overall sonic network is depicted.

5. EMERGENCE

In *Background Noise Study* both sound and music ‘emerge’ from a causal but nonlinear feedback system, from a network of signal-level interactions among components that either channel and transport sound energy, or amplify, reflect and filter it. The notion of *emergence* evoked here is not at all metaphorical. It has been adopted across various research

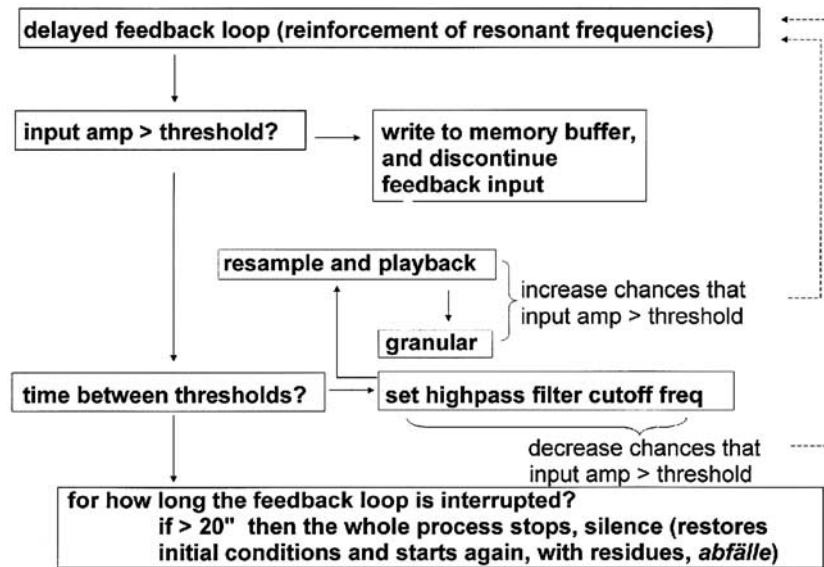


Figure 5.

domains (starting in the work of psychologist G.H. Lewes in 1875 (Mengal 2006)) to intuitively refer to the process by which a collection of interacting components shows collective behaviour (Baas and Emmeche 1997); in other words, higher-level properties of a *whole* are brought forth and sustained by several interconnected lower-level components mutually affecting each other.³

Visetti (2004) distinguishes between *synchronic* emergence from *diachronic* emergence. The first is the coming to existence of a process or entity based on a lower-level network dynamics. The latter is the articulation of that process or entity in time, through various morphologies, and subsequent stages and phases. As discussed in the two previous sections, *Background Noise Study* sets the conditions for sound to emerge from the barely audible ambient noise (synchronic), and for it develop or articulated over time (diachronic) in such a way that an elaborate sequence of sound events, of changing sonic shapes, is born.

Notice that emerged properties bind or constrain the lower-level processes they arise: they bias the local interactions and thus drive the system behaviour in its subsequent stages. Therefore, this is not just a bottom-up process, nor is it a top-down one. It implements a kind of distributed causality across levels: a complex system of *upward causation* (the overall technical infrastructure and the signal processing defined by the composer's own design) and *downward causation* (an emergent process of the upward causality that in

turn binds and determines further upward emergence, including downward causation itself as manifested at later stages). When *Background Noise Study* is performed, this two-ways causal bind is crucial for a sense of cause-and-effect perceptibly connecting the sound events, as well as for an overall direction in the resultant sound flow. Small details are susceptible to either augmenting/supporting or diminishing/inhibiting the potential of both the upward and the downward causation. A *deviation-amplifying mutual causality* is captured in the network of signal interactions, of a kind observed in complex dynamical across several research domains, including human sciences and economics (e.g. Maruyama 1963).

Emergence is not at all the same as the Gestalt principle *the whole is more than the sum of the parts*. It implies a co-dependency and co-determination of the parts, and a mutual connection of whole and parts, a more dialectical hierarchy: *the whole is often more than the sum of the parts, but it is often less than each individual part*. Once the whole has emerged, it can reduce or inhibit the action of specific individual components. Take, for instance, a large-scale social organisation: however complex, varied and efficient in its performance, the whole organisation can never be as rich and complex as each single human being participating in it. It induces simplified, typified behaviours that are functional to the whole, but detrimental to the individual.⁴

In the *Background Noise Study*, sonorities that could potentially emerge may remain just latent and hidden, never becoming actual. The strongest reso-

³Compositional and music-theoretical implications are discussed in Di Scipio (1994), where a *theory of sonological emergence* was proposed.

⁴I must have found that example in one of Edgar Morin's books, most probably Morin 1977.

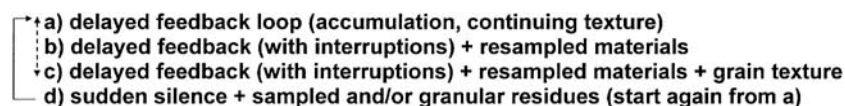


Figure 6.

nant frequencies in the room are usually expected to become among the prominent spectral features in the sounding result, yet in some presentations they have not prevailed at all over other, less obvious resonances, or have just remained silent. This does not mean that they had no effect: although not manifest to the ear, they still were *system resonances* influencing perhaps the level and onset time of other resonant peaks and spectral colourations. More generally, emergent properties can sometimes submerge and conceal other potential features (again an effect of downward causation). In such circumstances something still emerges, namely the *absence* of something that is expected to be there, something missing. Composition is not only a matter of bringing in and putting together – often it becomes a matter of letting out, if not one of prevention – the implication being that not all is possible, not anything goes.

6. FORM/PERFORM

In a more phenomenological description, a typical performance of *Background Noise Study* includes the following:

- a) the delayed feedback loop turns the feeble, dry and wide-band background noise into a spectrally more structured, sometimes reverberant and continuing texture;
- b) as (a) becomes louder, the internal samplers play back chunks of sound emerging from the feedback loop, altering their time-scale (frequency shifts) and adding more layers, thus augmenting the polyphony of the overall sound texture;
- c) the sampler output is in turn ‘granulated’ or ‘vaporised’, resulting in a weakening of the spectral focus that had been gradually reinforced through (a) and (b); energy gets dissipated across spectral regions, and dispersed both in time (low-grain densities) and space (grains are scattered across the speakers);
- d) if and when the process goes adrift, growing so much as to result in an excessively dense, loud and chaotically changing sound (saturation = lack of information renewal), it gets discontinued; as a more balanced situation is restored, it starts again from (a) (Figure 6).

Clearly, in an actual performance, things are not as linear as this description seems to have them. The process may often step back from (b) to (a) and from

(c) to (b). Sometimes it would wander randomly between (b) and (c) before ending with (d).

A full performance of *Background Noise Study* usually consists in going from (a) to (d) a number of times – say two, three or four, depending on various factors. In an installation, it would run indefinitely, so the gradual build-up and the eventual cut-up gesture would repeat many times.

As is clear from the discussion of ‘emergence’, the full-length contour or shape experienced in a concert performance of *Background Noise Study* – its ‘form’, if you like (I don’t) – is not at all formally represented or prescriptively determined. Nothing results from a scheduling of processes or events. The computer knows nothing of the particular events of sound and the overall, macro-level formal gesture. What one might perceive as discrete events – apparently arranged as sequences or as overlapping layers of sound – develop during the performance from the network of live interactions at the audio signal level. No scheduler, no sequencer exists. There is no declarative representation (event list), nor a procedural representation (algorithmic generation of control data and events) of the musical structure. All signal-processing algorithms are concurrent and synchronous. It operates of course in a determinate, *causal* way (not random). Yet, the sound source to work with is always different (each room has its own background noise, and a given room’s noise may be different at different times), and the room resonances affect and determine the unfolding of many details in the audio signal in a way that is far beyond deliberate determination. Sound is allowed to happen, and data are extracted from that sound and used to drive its extensions and modifications. Even gestures directly affecting the macro-level contour, like the cut-up following a more or less gradual drift, arise from processes at signal level.

Implicit in *Background Noise Study* is a statement that *music does not exist except as it is being made*; it never exists before it takes place, nor before you keep yourself listening to it. It welcomes the surrounding noise and makes something of it and with it. Its ‘form’ is the array of epiphenomena born of the performance process, the outcome of a well-balanced coordination among many system components including the room space, the equipment, and all people managing the equipment.

In this approach one thinks less about a pre-determined shape or contour (‘musical form’ normally understood) and more about the technical and

dynamical conditions that makes a sound process or system susceptible to being ‘formable’, to being ‘framable’ in the absence of pre-given frames. Thus one thinks of the *formability* or of the *formance* (if I may dare to use this term) of the materials emerging from the intricate network of sonic interactions. A ‘good’ or ‘efficient’ performance is when the several components go together well to actually make a system, a global entity whose behaviour is not reducible to the single parts or the separate local interactions. In this perspective, a composer does not compose the music, but composes interactions and interdependencies that manifest themselves as sound. These composed interactions make up a network, and the network is put in a real space, in a room, finding there resources to get started and live for some time.

An *ecosystem* dynamics is achieved when the array of networked interactions includes its *oikos* – the place where it is housed – as one of the systemic components. In our case, this is achieved via microphones and loudspeakers – in other words, transducers interfacing other system components, like signal processing algorithms, with the room sound. We have thus a computational activity that literally *takes place* in contact with the real, material sound environment, *in loco* and *cum loco* (in a place and with that place as an ineliminable system component). The emergence of sound and music happens with the immersion of the computational activity and the electroacoustic equipment in a real space, letting it go in real time. There is no shape or direction set beforehand, *there is no form without performance* (this is true in general except under the spell of a very idealistic mindset).

One point worth emphasising is the *fragility* of the system, its permanent exposures to risks of all sorts. *Performance* is really when the process succeeds in achieving a consistent behaviour in spite of this fragility. It needs exposure to the external space, but it also fears that very same exposure to. The risk is taken to fail in coming up with something that might be worthy of being called music. Definitions such as *live* electronics or *real time* are empty words if they don’t include the honest awareness that ‘every living existence is laden with a permanent risk of mistake and death’ (Morin 2004: 227, translation mine). For a living system (and for a live performance), no substantial abstraction is possible from the *hic et nunc* (here and now). The living requires a permanent exchange with the surrounding environment. Biologists and cyberneticians call it ‘structural coupling’ (Maturana and Varela 1980). However, the structural coupling is to be constantly negotiated and worked for.

How are we to evaluate a performance of *Background Noise Study*? What makes a performance ‘good’ or ‘bad’? The criteria involved may include principles of aesthetics, based on the sounding results obtained. As artists we are usually entitled to justify

ourselves in the name of aesthetic principles (beauty, symmetry, conceptual or emotional appeal, etc.). I personally don’t agree that aesthetics is the crucial point. A good performance of any of the *Audible Ecosystemics* works is when as many different system states as possible have been gone through, provided they are audibly revealed to the ear as different nuances of sound, as variations in texture, in timbre, in pace and density of gestures. A good performance is when a fairly large number of audible traces have emerged for the listeners to make sense of the coupling of equipment and room (i.e. the link between *oikos* and *system*) implemented and mediated by the people responsible for the technical (musical) process. A good performance is when the composed network of sonic interactions, although fragile and always at risk of failure, has preserved a sense of self-consistency and has, at the same time, remained subject to perturbations born of the material environment and the given working conditions (see section 9 below).

All notions of *form* imply *closure*, the possibility to draw a line and mark a distinction, the possibility to perceive differences and contrasts. *Formance* is a system’s availability to be closed. *Performance* is the process by which closure *takes place* (and that *takes time*). It consists in the sequence of interactions and exchanges bringing to a *conclusion*, i.e. bringing the system to close onto itself *with time* and *with space*.

7. OPEN/CLOSED (SPACE AS MEDIUM OF EXISTENCE)

In a performance of *Background Noise Study*, microphones and loudspeakers are not primarily to capture and reproduce sound: their main task is to generate sound and capture traces of the surrounding space. They become *musical* instruments (Mulder 2010).⁵ More precisely, they become *system terminals*, interfacing the computational activity with the room space. They are the very interfaces by which the overall process can do something to the space and can be affected by the space response (and by independent events happening in the space). Therefore, they are essentially the *loci* of the structural, permanent coupling of machine, human and room. By means of sound terminals, *Background Noise Study* does something to the room, and has something done in return by the room. Changing or removing a system terminal would be like killing the process, or at least it would be like modifying it so much as to change its system identity, its unique profile as an

⁵Of course, the direct use of transducers as musical instruments has a long and relevant tradition in experimental electroacoustic music, especially vivid in approaches of circuit bending and hand-made electronics (Collins 2006).

entity operating with specific interrelationships and interdependencies of component parts.

In the performance, the dynamical process keeps itself always open to the surrounding space. This fundamental openness allows the system to 'hear' – and thus to be affected by – sound events in/from that space. The latter include sounds born of reasons independent of the performance itself (someone coughs, the lamps buzz, the door creaks, the fan hums...), but also include the sound materials generated by the work itself in its process, diffused by the loudspeakers in the room. The system's sound always overlaps with the room response. There is no way to isolate the system's 'own' sound from the room's ambience. *There is no way to isolate the system input from its own output, as all output is an input.* The very idea of an input/output system should be abandoned, in this context. *The room space becomes the medium through which the process hears itself and acts upon itself.*

Thus the system is *open to the room*, but this openness is the same as being *closed onto itself through the room*. The system hears itself unfolding, constantly affected by the material, physical characteristics of the room (the latter of course are nothing 'natural'; they stem from culturally connoted and socially functional designs). The structural coupling of a given process or system with the space surrounding it makes the space a part of the system; it inserts the space – i.e. the *oikos*, the house place where life unfolds – into the system, and reveals it as a *medium of existence* (a notion drawn from Maturana and Varela 1980). Without a room, without terminals branching in the room, *Background Noise Study* would just stay silent, would be nonexistent. In an ecosystemic perspective, 'space' is never something outside the intended system, like an external factor. We will address shortly the notion that, in the medium of sound, nothing is purely external just as nothing is purely internal.

A variant work, *Background Noise Study with Mouth Performer* (also known as *Background Noise Study, in the Vocal Tract*) develops a kind of audible dialectics between two rooms: a larger room, housing the performance, and a much smaller one – a small niche within the larger room – represented by the mouth cavity of a performer. In a more recent installation, *Untitled Ecosystemic Installation in Dismantled and Abandoned Rooms* (Berlin, 2008), two or more rooms are coupled and mutually influence one another's ambient sound.

Such works question the notion of separate, isolated spaces. They question the idea according to which a space can be represented within another space (i.e. the 'spatialisation' typical of most electro-acoustic music). They provide occasions for auditive experiences which stand in sharp contrast with talks and dreams of *virtual spaces*. The space experienced

here is material, physical, and architecturally and socially connoted.

8. ECOSYSTEM DESIGN/RADICAL CONSTRUCTIVISM

Sound ecosystems are obviously artificial entities: no living system exists purely in the medium of sound. A work like *Background Noise Study* calls for no naturalistic connotations. It stems from no scientific modelling of extrasonic phenomena. The idea that an ecosystemic dynamics can be designed and made to operate, here-and-now, in time- and site-specific conditions, implies a theoretical challenge of both an artistic and bioengineering nature (Prominski 2007).

We *construct* a nature; we do not replicate or model a segment of extant nature. This approach leans on principles rooted in the cybernetics of the living (Maturana and Varela 1980), neurophenomenology (Varela 1992), and ecosystem theory (Jørgensen and Müller 2000). However, it is never intended to sonically replicate an objectively observed reality (for that matter, none of the mentioned authors would agree that something like an objective reality does exist). The approach of the *Audible Ecosystemics* project stems from a radical constructivistic view.⁶ The 'real world' is one's own construction (including knowledge, and sharing, of other human beings' own constructions).

9. IDENTITY/CHANGE

Closure – the notion of a sound-generating and self-regulating system closed onto itself through the surrounding space – implements *identity*. In a performance of *Background Noise Study*, the network of sonic interactions constantly strives to preserve its basic closure; it works in such a way that no foreign events, however disruptive and unexpected, would modify its system functions. (Note that, thanks to signal processing functionalities not described here, excessively loud room sounds born of, say, aggressive sources are automatically discarded: the process tries to protect itself from arrogant agencies stepping across its vital space.)

The closure of the system onto itself through the room implies a recursive transfer function. If we were to mathematically model the whole thing, we would come up with a nonlinear iterated function system. Yet, as stated already, *Background Noise Study* does not stem from a formal, mathematical model at all.⁷ It is a concretely operating hybrid system.

⁶See von Ameln (2004), especially sections on Heinz von Foerster and Ernst von Glasersfeld.

⁷For me, this is no longer the time for formalised models, however innovative, generative and complex they can be.

Its interactions with itself are nonlinearly mediated through the room via electroacoustic transducer terminals, and are designed as basic digital signal processing methods.

Openess – the notion of a sound-generating and self-regulating system that seeks in the surrounding resonant space the resources to live on and with – implements *change*. It is an exposure to perturbations that makes it react and adapt, changing the way it manifests itself. In a way reminiscent of living systems, including ecosystems, the *Background Noise Study* process is *structurally closed but operationally open*. It strives to keep itself working (it strives to establish and preserve its self) by exposing itself to new solicitations and responses from the room (by questioning its self). Identity is thus flexibly redefined as a function of acceptable, nondestructive patterns of interactions.

It may be surprising to note that such *systemic identity* is precisely what musicians usually call *timbre*: the array of traces of given physical constraints, the audible signature of interconnected physical forces acted upon by a human agent, subject to particular solicitations from the environment. Now, *space* – as a necessary medium of existence for all sound phenomena – can be experienced as the medium where interactions of interconnected physical forces take place to build an identity (timbre).

As mentioned above, in live sound and live space, everything is connected with every other thing. A strong statement in *Background Noise Study*, and other installations of mine, is *that no sound as such ever exists*. All sound is, to some greater or smaller extent, the audible trace of a clash or encounter of physical bodies that are either moving and reflecting, or touching and altering one another. The identity of a sound (and, at a different scale, the identity of a sound-generating computational activity such as the *Background Noise Study*) is the same with a necessary exposure to otherness. An audible *self* is sought as a performance proceeds: sonic identity (form, timbre) is the self-consistent manner by which a system changes upon meeting with what appears as *non-self* (this is my way of capturing Francisco Varela's profound understanding of the self of living systems (Varela 1991)).

10. EVENT/OBJECT (POLITICS OF SOUND)

In the performance of *Background Noise Study*, the loudspeakers are turned against the room walls. That is to stress that no sound 'as such' is intended, no musical message that might be separated by the transmission channels and by the material context of listening. Most of what is there to be heard is reflected and indirect (mediated), and depends on contingent, ineliminable and irreplaceable factors. A live performance of *Background Noise Study* could

be recorded (Di Scipio 2005a), but listening to the recording returns only a pale image of the work: the particular work in fact declares listeners as *part* of its live performance – as opposed to *observers standing in front of it*.

I think today we need to state very clearly that *sound* should not be confused with, or reduced to, *sound object*. The latter term refers of course to a long-standing tradition, born of Pierre Schaeffer's early *musique concrète*. In the meantime, due to cultural and economic processes beyond the scope of this article, it has become a pervasive concept in all (electronically mediated) music: we all too often say 'sound' and mean 'sound object'. We should state very clearly that *sound* is always *event* and never object. It is energetic, ephemeral and informational: it is nothing that can exist 'in itself' or 'as such'; it is the traces left by physical mediations and interactions.

Recording technologies are substantive to an understanding of sound as object. Yet even recorded sound is not entirely an object 'in itself', as it shows traces at least of the specific audio equipment (human competences and designs) and the rooms (social configurations) by which it is conveyed to ears. Listening is essentially addressing oneself to sound *events*: it implies a welcoming of, and a caring for, something audibly happening here-and-now. Listeners always take part in the event of sound they listen to. To a greater or smaller extent, they modify it and are (bodily, cognitively, emotionally) modified by it. A manifest implication of the recent 'ecosystemic sound construction' entitled *Stanze Private* (Private Rooms, 2008–09) is that visitors never attend to the work 'as such', and always attend to the work as it is modified by their physical presence. The listener (observer) changes the listened to (observed): echos of the Heisenberg indeterminacy principle.

The notion of sound object (sound captured, stored, replicated, analysed, visualised) has long seemed perfectly appropriate to music 'in the Age of Mechanical Reproduction' (to echo Walter Benjamin's 1936 essay (Benjamin 1955)). Many tend to think that the reproduction technologies are neutral to sound (objects), not interfering with the meaning and structure of the music they convey and mediate. In contrast, most sound installation art is not reproducible by recording media, no matter how many audio channels can be utilised. The implicit emphasis on the here-and-now, it could be suggested, imbues the work of art with the *aura* that mechanical reproduction had destroyed: Benjamin's theory of media, too often unproblematically applied by scholars and musicians to music and the sound arts, may be revealed simply inadequate or obsolete.⁸

⁸See Mersch 2002 and R  th 2008 for introductory remarks on this important point.

Like other artists in the world, I pursue a perspective contesting the notion that audio technology is only a means of re-presentation. It also resists a notion that music is made of discrete sonorous items, like separate objects each having a definite profile of its own, unaffected by the other objects, by listeners, by the surrounding space, by the equipment.

This latter observation points to an issue that is quite problematic for critics and musicologists. Solomos and Meric (2008) ask: how are we to analyse a work such as *Background Noise Study*? In a way, it resists *analysis* – that is, it resists strategies of breaking down the whole into component parts that can be independently characterised. For me, that is more than a musicological question. ‘Resisting analysis’ may in fact support a sense of *solidarity*, a sense of staying together and belonging, of mutual exchange and systemic binding. Related issues, particularly issues concerning the technologies involved in the performance ecosystem, would also bring us to social matters – technologies of course being mediators of human relationships and production forces (see Green 2008). As Solomos (2008) observes, social and political issues are *enacted* (they are not *represented*) in the process by which *Background Noise Study* starts sounding and develops in time. This is because it has much more to do with the enactment of a sound world than with the representation of a sound world.

11. HUMAN AGENCIES

Listeners are not external to the music. On the one hand, they are addressed by the performance not as individuals, but collectively: the unity of time and place specific to the work turns separate listeners into a group, which we call usually an *audience*.

On the other hand, the way the members of the audience sit or otherwise occupy the room is not without consequences on the room acoustics; the ambient sound, with the audience in the room, is not the same as the ambient sound without the audience. The *audience* is the *ambience listening*.

All listening comes with a physical presence, with bodies, with sounds. Listeners are expected to stay rather silent, in some social contexts. But even listening in silence is not entirely without sound. The audience has the sonic presence of a small community made of individuals sharing a time and a place for paying attention to some sounds. Their sonic presence becomes an integral part of the *Background Noise Study* performance, because it is to be considered as part of the room background noise.

The person or team in charge of the performance is expected to find a secure but dynamical balance in the exchange between the equipment and the room acoustics. Preparing the performance is for them a matter of trying different microphones and locating the precise



Figure 7. Natalia Pschenitschnikova, ‘mouth performer’, in Agostino di Scipio’s *Background Noise Study, in the Vocal Tract* (Weimar, January 2008)

spots where microphones should stand; of trying different placements and orientations for the loudspeakers; of developing a sense for the background noise specific to the given room. Many other factors could be mentioned. The person or team in charge of the performance should be acknowledged as having full interpretative responsibility. Their task is not to re-present and correctly convey something already there, prepared beforehand. Their task is to make actual and present something that, prior to their own action, remains purely latent and potential. During the performance, they possibly just listen, thus becoming members of the audience. However, they can still take action and function as a complex adaptive mediator between other system components. Responsible human action is decisive for an acceptable balance to be established between man-made artefacts (equipment) and environments, either natural or man-made (room).

In *Background Noise Study, in the Vocal Tract*, a very special instance of human agency is introduced as an integral part of the delayed feedback loop. A performer has to handle a miniature microphone in her or his mouth (Figure 7). For the most part, the performer stays rather still, doing nothing except holding the miniature microphone in the mouth, with the fingers, and changing mouth postures (causing changes in the resonant structure of the feedback loop). Sometimes she or her takes a rest, and takes the small microphone out the mouth. Then, the microphone gets the room sound, too, so the room sound unfiltered by the mouth will eventually overlap with the vocal resonances in the delayed feedback loop output.

‘Staying still’, ‘doing nothing’, not even breathing (or breathing through the nose cavities, letting no air into the mouth); keeping mouth and lips in a specific vowel posture for some time, then smoothly changing vowel, every now and then – all this is not without sonic by-products. With time, the mouth muscles get tired, so

keeping the posture makes them gradually more tense: saliva droplets are formed; the tongue sometimes moves, with a tiny water-like sound; the jaw bones too sometimes move and make sound, as the mouth and lips change vowel posture from, say, /i/ to /u/; every now and then the performer needs to swallow and to moisten the lips; not breathing (or breathing only through the nose) is not always perfectly successful. In short, the control exerted over muscles and bones may dwindle, making unwanted but necessary sounds.

These unwanted noise events become part of the main feedback loop, just like the room noise. Relatively louder events will eventually result in a chain of cause and effect making the total sound become denser and more chaotic, until the total process grows too much and goes adrift, apparently out of control.

Facing that *emergency situation*, the vocal tract performer takes one or more *security measures*: she or he is then allowed to deliberately produce some sound (humming a note, breathing into the microphone, etc.), which causes the microphone input level to grow much louder relative to the background noise. Due to the compensative criteria in the signal processing (described in a previous section), the feedback loop will eventually be discontinued, so the sonic drift is interrupted, or at least settled. The process then restarts in a more balanced situation, if not in silence, except for possible waste sounds left by the drift and possible by-products of the security measures taken.

The body (mouth, lips, tongue) of the human performer constitutes, in such a context, an integral part of the ecosystem dynamics. For listeners, there is a sense that everything depends on the most tiny movements in the mouth and the way they resonate in the room when amplified through the loudspeakers: a sense of uncertain and precarious balance. The performer's most minute actions shape the sound process and the audible ecosystem. At the same time, her or his actions are driven by what she or he hears and perceives of the surrounding sound. In a way, then, the audible ecosystem enacts itself thorough the decisions and actions taken by the human agency.

Humans make and define their ecosystem inasmuch they are made and defined by the ecosystem. They are responsible for the ecosystem, inasmuch the ecosystem is responsible for their life and development. Humans are responsible for themselves through the ecosystem. The ecosystem is responsible for itself through human beings' actions in it. Such is the position of the performer in *Background Noise Study, in the Vocal Tract*. Again, no explicit political or ecological statements are spoken in the piece. Yet politically and ecologically relevant principles are active, as they implement the process that makes sound happen and change in time in a way possibly worthy of calling music.

The taking of security measures – actions meant to cope with an emergency situation – implies a decision

to exert a little violence. In our case, a little violence is made to the system process with a specific goal in mind: a more balanced and stable sound texture, or even silence. The way by which security measures are actually implemented is important. A smooth and tolerant action would be successful, but only over a longer time: the goal is eventually achieved, but it takes a while. A more aggressive action would achieve the goal much sooner, yet it might cause undesirable reactions and thus give rise to unexpected consequences: it certainly settles the situation, but next time around, the sound drift could grow denser and thicker, more agitated, harder to settle (remember that such behaviours, described here in narrative terms, are emergent features of the real-time signal-level process).

The sounds born of the performer's dealing with an emergency, audibly reflect her or his way of coping with circumstances that are critical for the ecosystem balance. The personal responsibility can be listened to in terms of the musicality of the resultant sound gesture, and it may be listened to in terms of the willingness of the performer to face the consequences of her or his own actions (wanted or not).

12. FURTHER WORK

The work described here raises issues of both particular and, in my view, more general relevance to current artistic practices. I have tried to illustrate some of them, taking a work like *Background Noise Study* as a representative example. However, the list of themes and key words I have actually discussed is far from complete. A few important issues have been simply omitted, despite their significant implications. Interested may readers find more in Di Scipio (2003, 2005b, 2008).

For example, notions such as *feedback* (positive and negative) and *interaction* are often taken to be very well-known, even commonplace notions. Yet they would actually require much deeper insight, especially in that they are rooted in system-theory and ecology-related ideas today foreign to our daily use of such terms. Besides, we tend to forget the political resonances they do have.

Talking of human agency in relationship to technologies and the environment, the human body itself – the body of the performer, as well as the body of the listener/viewer – is a subject that would need deeper investigation.

A particular question concerns the role and meaning of 'scores' as relative to the kind of practice presented here. What kind of documentation represents the score for a work such as *Background Noise Study*?⁹ Is it 'recipe' or 'instruction manual'? In general,

⁹Most of the works mentioned above have 'scores' and documentation materials, available from <http://xoomer.virgilio.it/adiscipi/materials.htm>

do live-electronic works and sound-installation works need scores in order to be properly circulated? What do they gain from being replicated and possibly re-interpreted? To what extent can the particular technological processes and functions be shared with other practitioners? I believe details such as digital signal processing flow-charts, electroacoustic chain graphs, and so on should be illustrated in ways independent of given equipment and specific software tools, but that again raises questions of notation.

I think that questions such as those I have discussed here, from my own personal perspective, are to be spoken about by a larger number of prospective readers and practitioners. As themes reaching far beyond individual experiments and research, they should be the subject of wider discussions and collective efforts, hopefully becoming the catalyst of practical and theoretical contributions, making the very necessary link between creative efforts in sound and broader social and political questions. The present issue of *Organised Sound* certainly represents a valuable step in that direction.

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DEDICATION

For A.

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